

Catherine Yu-Chi Chen

Ph.D. Candidate, Computational and Mathematical Engineering

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Education

Stanford University

Ph.D., Institute for Computational and Mathematical Engineering.

Stanford, CA
09/2022 - 08/2027 (Expected)

Columbia University

B.A., Data Science. GPA: 3.95 (Cum Laude)

New York, NY
09/2018 - 05/2022

Awards

- ICME Instructor Teaching Award 2025

Fellowships

- Professor Joseph E. Oliger Memorial Graduate Fellowship 2026 - Present

Research

Stanford University

Research Assistant to Assistant Prof. Lihua Lei

Stanford, CA
12/2023 - Present

- **Conformal Tail Risk Control:** developed distribution-free methods to control empirical tail risk for black-box models; contributed theoretical analysis and experimental evaluation for LLM alignment settings
- **Online Conditional Value-at-Risk (CVaR) Control:** implemented and evaluated an online CVaR control algorithm based on the Rockafellar-Uryasev formulation and parameter-free online learning; ran large-scale simulations under adversarial and non-stationary dynamics for LLM toxicity control and portfolio management task
- **Inflation Forecasting via Microdata:** built forecasting models using micro-price data and macro covariates via SARIMA, XGBoost, and Random Forest; designed rolling-window and expanding-window validation and error analysis for long-horizon forecasting
- **Synthetic Control with Interference:** designed a procedure for conducting synthetic control under the influence of spillover effects

Columbia University, Center for Theoretical Neuroscience

Research Assistant to Prof. Liam Paninski

New York, NY
09/2021 - 06/2022

- Built a **semi-supervised sequence model** for behavioral segmentation on large-scale mouse video data (100k+ frames), implementing the full training and evaluation pipeline

Columbia University, Digital Video and Multimedia (DVMM) Lab

Research Assistant to Prof. Shih-Fu Chang

New York, NY
09/2020 - 08/2021

- Built **multimodal NLP and computer vision pipelines** for sentiment and event analysis on city-scale social media data supporting NYC Mayor's Office analytics

Publications

- C. Chen, C. Gao, C. Lian, J. Hazell, L. Lei. Inflation Forecasting with Microdata. *Work in progress*.
- C. Chen, S. Shen, Z. Deng, L. Lei. (2026) Adversarially Robust Control of Conditional Value-at-Risk via Kelly Conformal Inference. *Proceedings of the 43rd International Conference on Machine Learning (ICML)*
- Chen, C., Shen, J., Deng, Z., Lei, L. (2025) Conformal Tail Risk Control for Large Language Model Alignment. In *Proceedings of the 42th International Conference on Machine Learning (ICML)*.

Presentations

Adversarially Robust Control of Conditional Value-at-Risk via Kelly Conformal Inference

- INFORMS Annual Meeting (10/2025)
- Eastern North American Region (ENAR) (03/2026)
- Stanford Causal Science Center Conference on Experimentation (05/2026)
- International Conference on Machine Learning (ICML) (07/2026)

Conformal Tail Risk Control for Large Language Model Alignment

- INFORMS Annual Meeting (10/2024)
- Stanford Causal Science Center Conference on Experimentation (05/2024)
- International Conference on Machine Learning (ICML) (07/2025)
- Annual Conference on Neural Information Processing Systems (NeurIPS) (12/2025)

Teaching

Instructor

- CME 300Q: ICME Qualifying Exams Workshop, (Stanford University, Spring 2024, Spring 2025)
- CME 192: Introduction to MATLAB, (Stanford University, Winter 2025)

Teaching Assistant

- CME 241: Reinforcement Learning with Applications in Finance, (Stanford University, Winter 2026)
- MGTECON 614: Emerging Topic in Econometrics, (Stanford University, Fall 2024, Spring 2026)
- CME 291: Master's Research, (Stanford University, Fall 2024)
- CME 102: Ordinary Differential Equations, (Stanford University, Fall 2023)
- MATH 1102: Multivariate Calculus, (Columbia University, Fall 2021)

Other Academic Services

- Vice President of Stanford's Society of Industrial and Applied Mathematics (SIAM) (2023 - Present)
- Secretary in Stanford's Women in Mathematics, Statistics, and Computational Engineering (2023 - Present)
- Member in the Institute for Computational and Mathematical Engineering Student Activity Group (2023 - Present)
- Mentor in Stanford's Women in Math Mentoring (2023 - 2025)

Industry Experience

Apple, Exploratory Design Group (XDG)

Mathematical Sciences Intern to Jeff Koller

Cupertino, CA

06/2024 - 09/2024

- Prototyped **uncertainty quantification and missing-data inference pipelines** for internal forecasting and decision-support systems
- Built a simulation and evaluation framework for **imputation methods** under structured missingness; produced internal benchmarks

Google

Software Product Sprint: Machine Learning Participant

Taipei, Taiwan

07/2020 - 08/2020

- Designed and generated a book recommendation system using machine learning algorithms

Point Robotic Medtech Inc.

Software Engineering Intern

Taipei, Taiwan

07/2019 - 08/2019

- Implemented **3D reconstruction pipeline from 2D medical images** for surgical robotics prototype systems